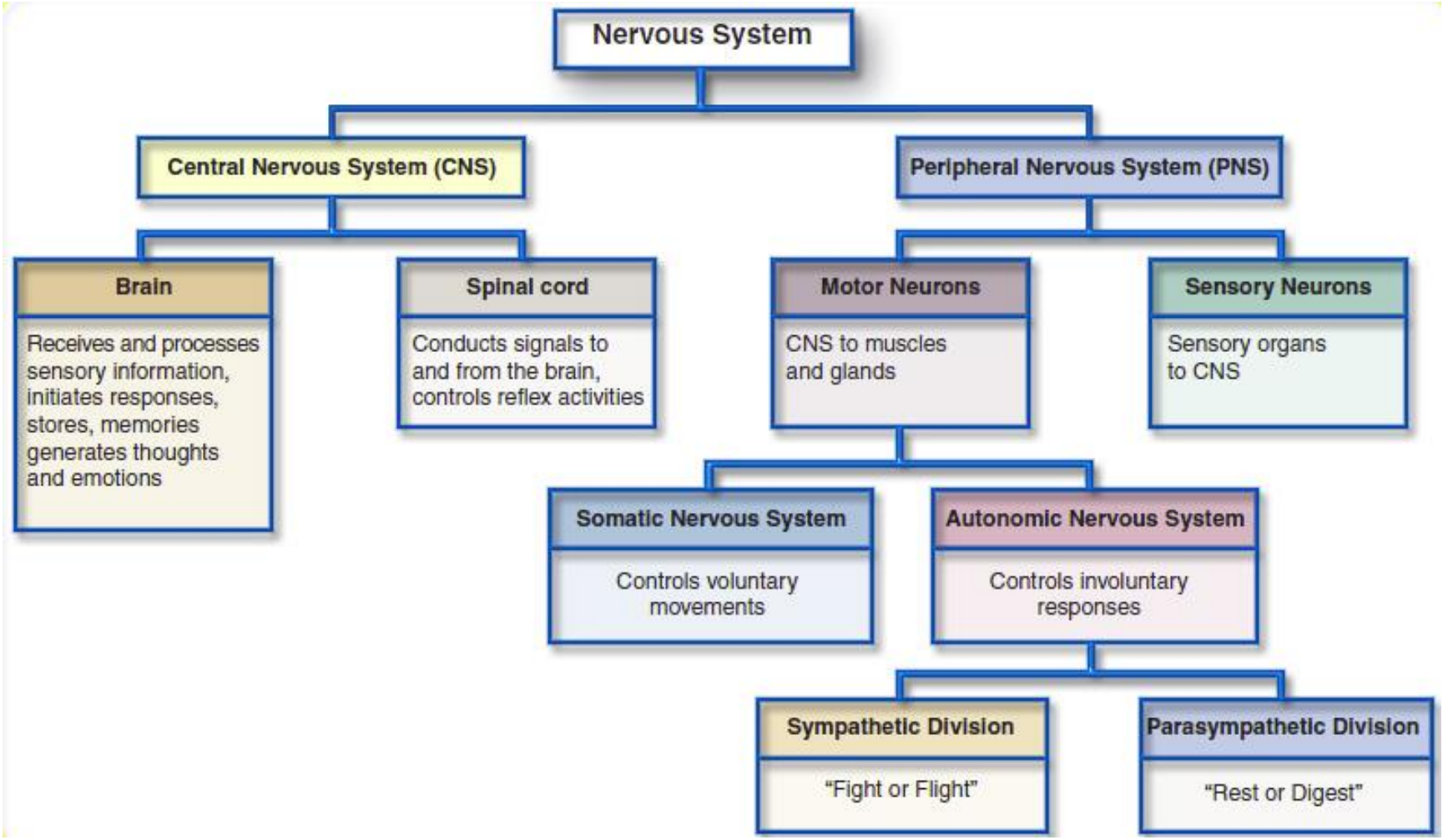
Introduction to Autonomic nervous system



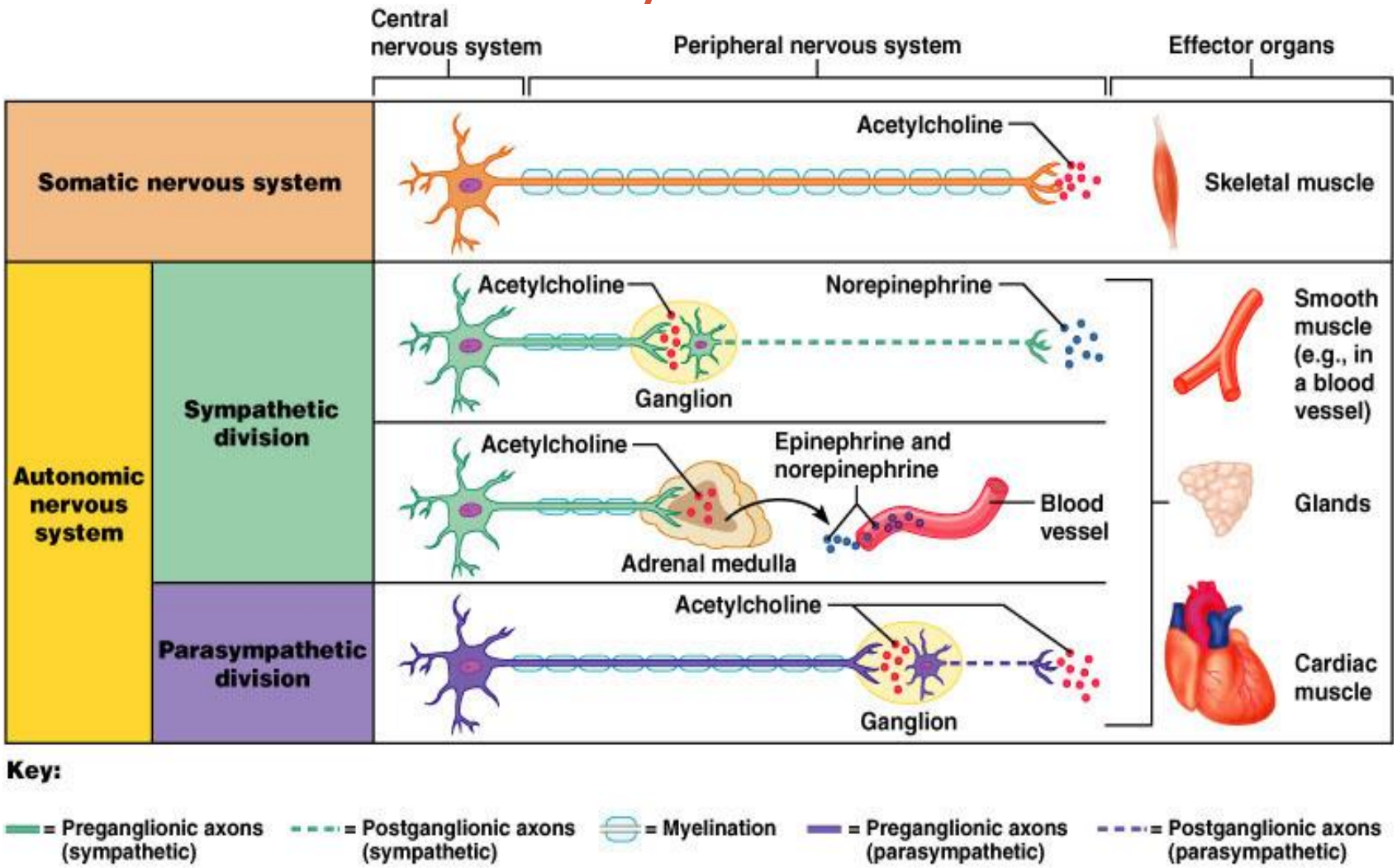
Nervous System



Peripheral Nervous System (PNS)

1. Somatic branch
   - Nerves innervate skeletal muscle; Voluntary
2. Autonomic branch
   - Controls internal organs;
   - Generally involuntary
   - Innervates the following:
     - **Smooth muscle**
     - **Cardiac muscle**
     - **Vasculature**
     - **Exocrine glands**

Comparison of Somatic and Autonomic Systems



Anatomical Differences in Sympathetic and Parasympathetic Divisions

Come from different regions of the CNS

- **Sympathetic**—from the **thoracolumbar** region
- **Parasympathetic**—from the **craniosacral** region

Differing locations of ganglia

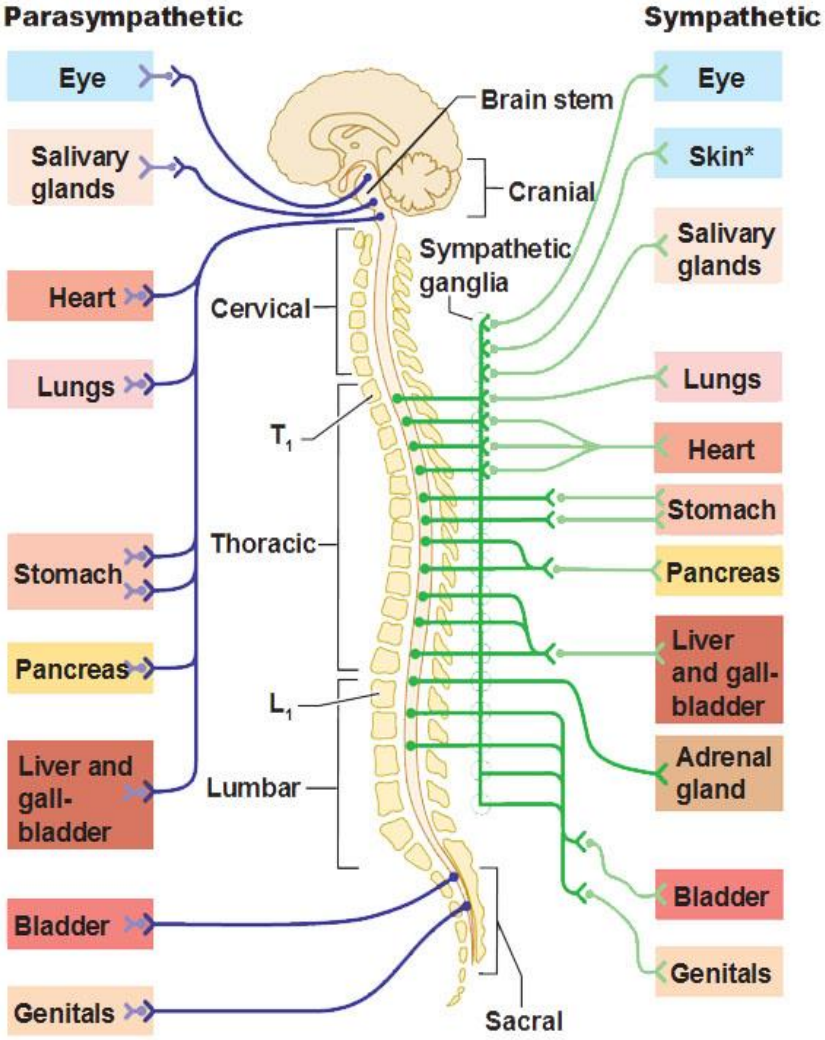
- Sympathetic – close to spinal cord in a chain
- Parasympathetic – close to target organs

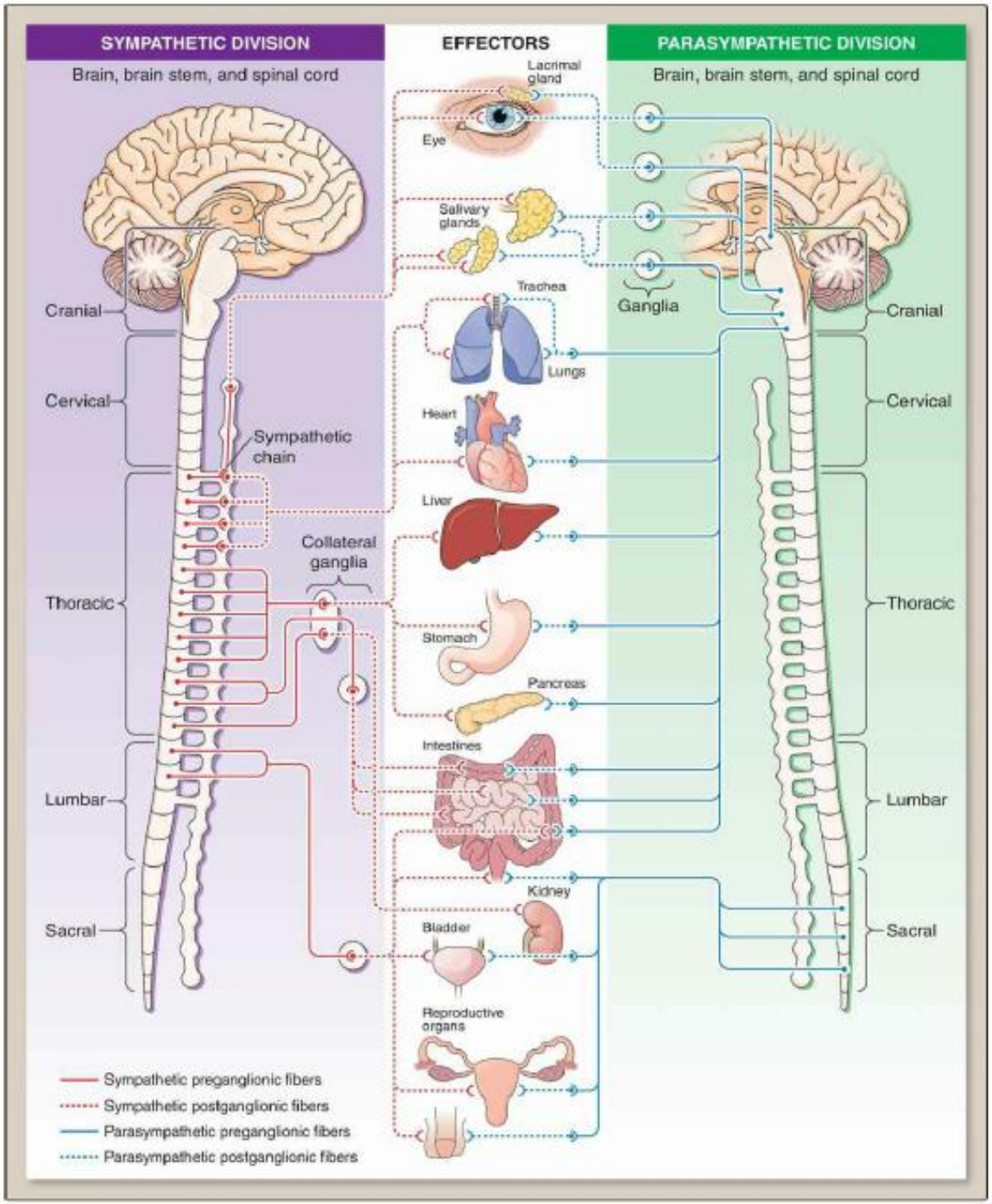
Differing lengths of *postganglionic* fibers

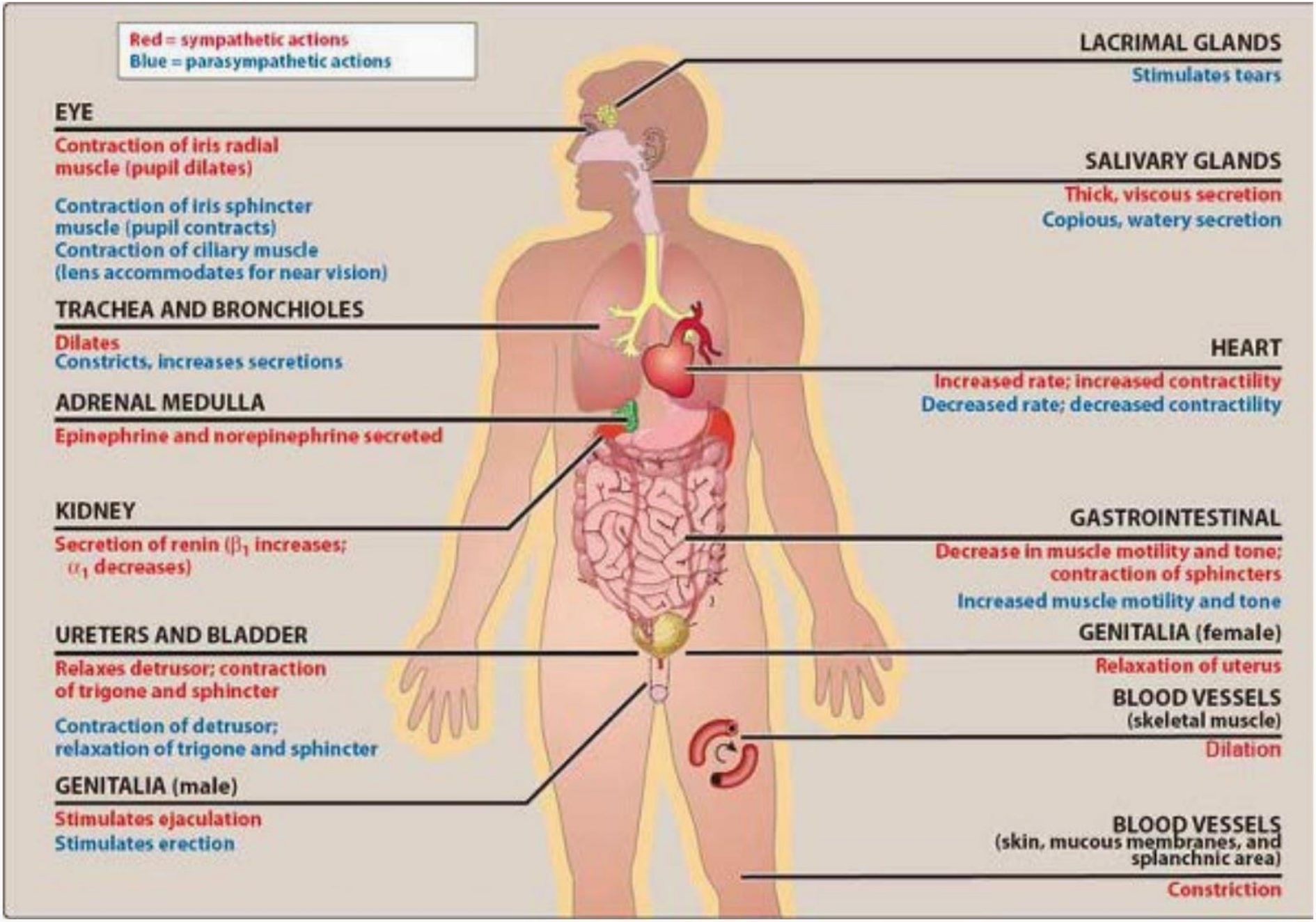
- Sympathetic – Long
- Parasympathetic – Short

Postganglionic branching

- Sympathetic – lots, so that multiple organs can be mobilized at once
- Parasympathetic – very little branching




SYMPATHETIC DIVISION
Brain, brain stem, and spinal cord
EFFECTORS
PARASYMPATHETIC DIVISION
Brain, brain stem, and spinal cord
Lacrimal gland
Eye
Salivary glands
Trachea
Lungs
Heart
Liver
Stomach
Pancreas
Intestines
Kidney
Bladder
Reproductive organs
Ganglia
Cranial
Cervical
Thoracic
Lumbar
Sacral
Sympathetic chain
Collateral ganglia
Sympathetic preganglionic fibers
Sympathetic postganglionic fibers
Parasympathetic preganglionic fibers
Parasympathetic postganglionic fibers


Red = sympathetic actions
Blue = parasympathetic actions
EYE
Contraction of iris radial muscle (pupil dilates)
Contraction of iris sphincter muscle (pupil contracts)
Contraction of ciliary muscle (lens accommodates for near vision)
TRACHEA AND BRONCHIOLES
Dilates
Constricts, increases secretions
ADRENAL MEDULLA
Epinephrine and norepinephrine secreted
KIDNEY
Secretion of renin (β_1 increases; α_1 decreases)
URETERS AND BLADDER
Relaxes detrusor; contraction of trigone and sphincter
Contraction of detrusor; relaxation of trigone and sphincter
GENITALIA (male)
Stimulates ejaculation
Stimulates erection
LACRIMAL GLANDS
Stimulates tears
SALIVARY GLANDS
Thick, viscous secretion
Copious, watery secretion
HEART
Increased rate; increased contractility
Decreased rate; decreased contractility
GASTROINTESTINAL
Decrease in muscle motility and tone; contraction of sphincters
Increased muscle motility and tone
GENITALIA (female)
Relaxation of uterus
BLOOD VESSELS (skeletal muscle)
Dilation
BLOOD VESSELS (skin, mucous membranes, and splanchnic area)
Constriction

Sympathetic Reaction: Fight or Flight

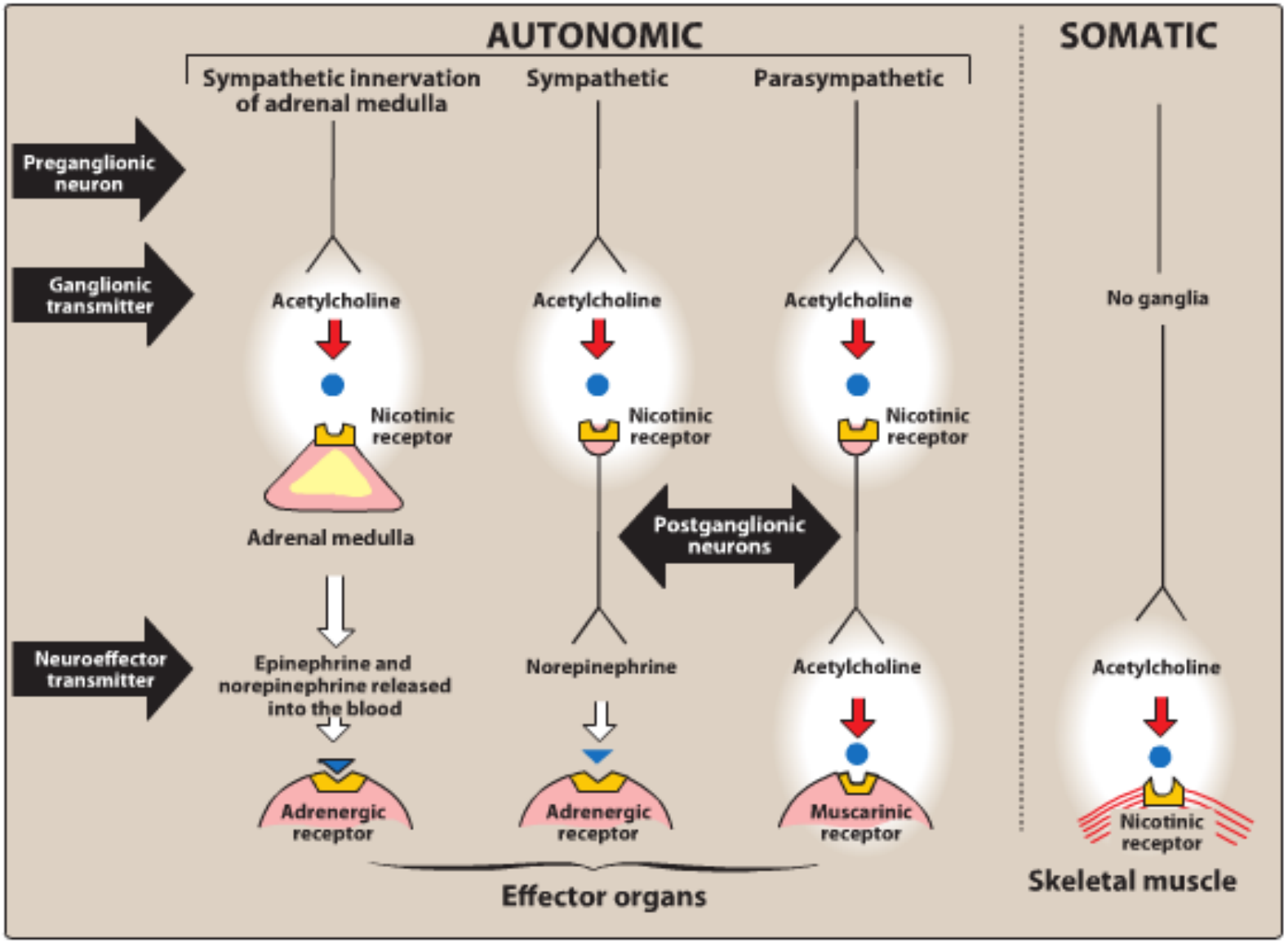
- During rage or fright the sympathetic system can discharge as a unit--affecting multi-organ systems.
  - Increase heart rate
  - Increase blood pressure
  - Increase blood flow to skeletal muscles and heart
  - Increase blood glucose
  - Bronchioles dilatation
  - Pupils dilatation: contraction of iris radial muscle
- Many of these effects are primarily due to epinephrine release from the adrenal medulla.

Parasympathetic Responses: Rest and Digest

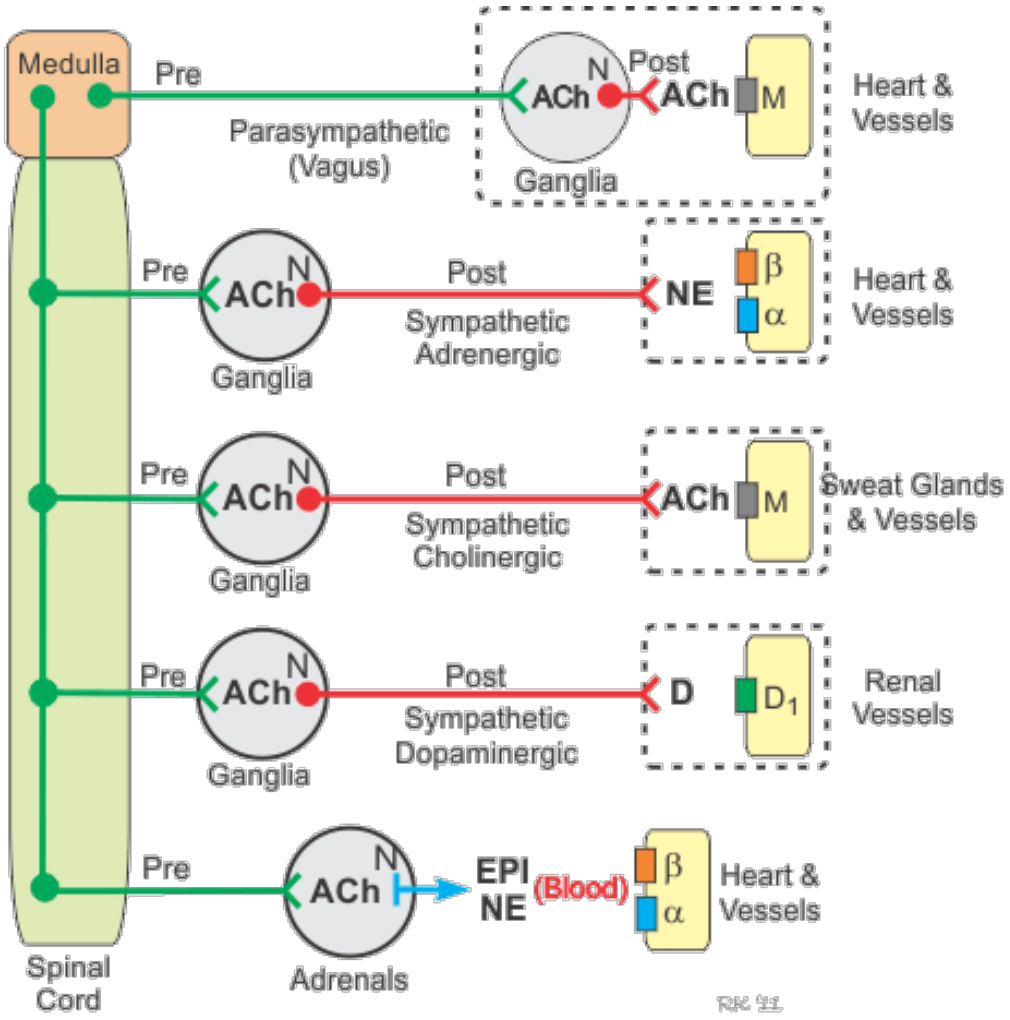
- Parasympathetic system controls more discretely, reducing the level of resting organ function.
  - Slows heart rate
  - Lowers blood pressure
  - Increases gastrointestinal motility
  - Protects retina from excessive light (Pupil: constricts): contraction iris sphincter
  - Empties the bowel and bladder
  - Promotes absorption of nutrients

Innervation: General Characteristics

1. Most organs are innervated by both branches
2. When both innervate an organ, they usually oppose one another
3. One system usually predominant in controlling the activity (e.g;. Vagus nerve is the predominant factor for controlling heart rate)
4. Some effector organs (adrenal medulla, kidney, pilomotor muscles, and sweat gland) receive only sympathetic innervation


AUTONOMIC
SOMATIC
Sympathetic innervation of adrenal medulla
Sympathetic
Parasympathetic
Preganglionic neuron
Ganglionic transmitter
Neuroeffector transmitter
Acetylcholine
Acetylcholine
Acetylcholine
No ganglia
Nicotinic receptor
Nicotinic receptor
Nicotinic receptor
Adrenal medulla
Postganglionic neurons
Epinephrine and norepinephrine released into the blood
Norepinephrine
Acetylcholine
Acetylcholine
Adrenergic receptor
Adrenergic receptor
Muscarinic receptor
Nicotinic receptor
Effector organs
Skeletal muscle

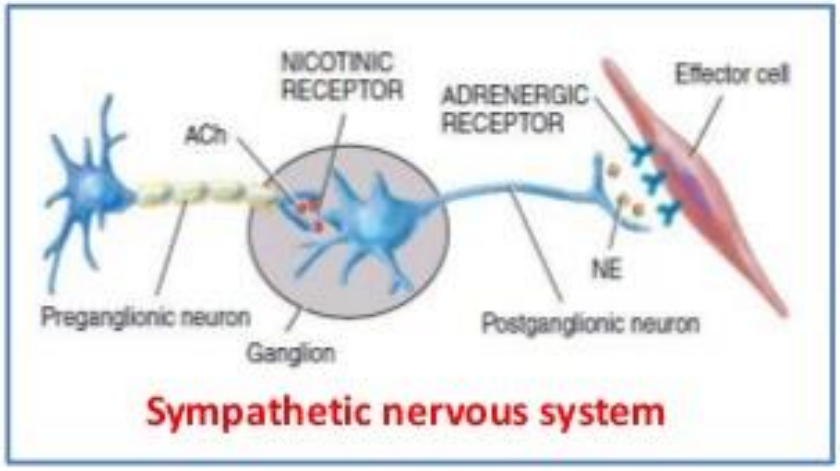
ANS neurotransmitters and receptors

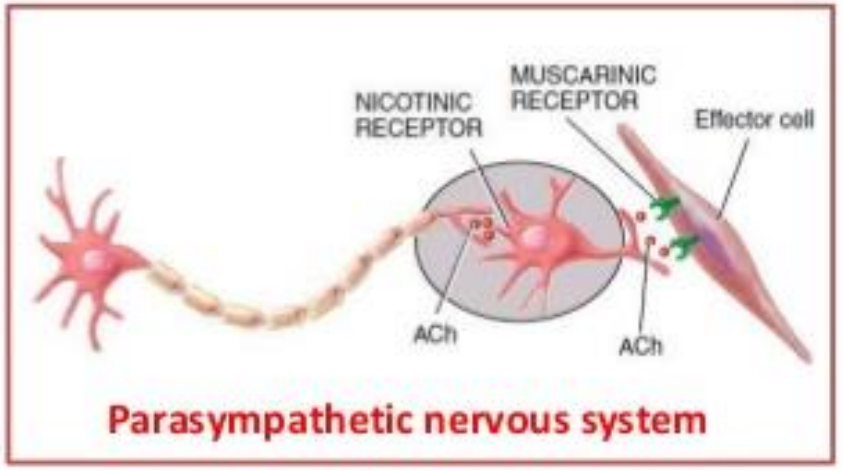


CNS = central nervous system; Pre = preganglionic; Post = postganglionic;
ACh = acetylcholine; N = nicotinic receptor; NE = norepinephrine; EPI = epinephrine;
D = dopamine; M = muscarinic receptor; β = β-adrenoceptor; α = α-adrenoceptor;
D_1 = dopaminergic receptor

ANS neurotransmitters and receptors

ANS	Receptor	Receptor Sub-type
Parasympathetic nervous system	Nicotinic cholinergic receptors	Nn, Nm
	Muscarinic cholinergic receptors	M1, M2, M3, M4, M5
Sympathetic nervous system	α adrenergic receptor	α1, α2
	β adrenergic receptor	β1, β2, β3



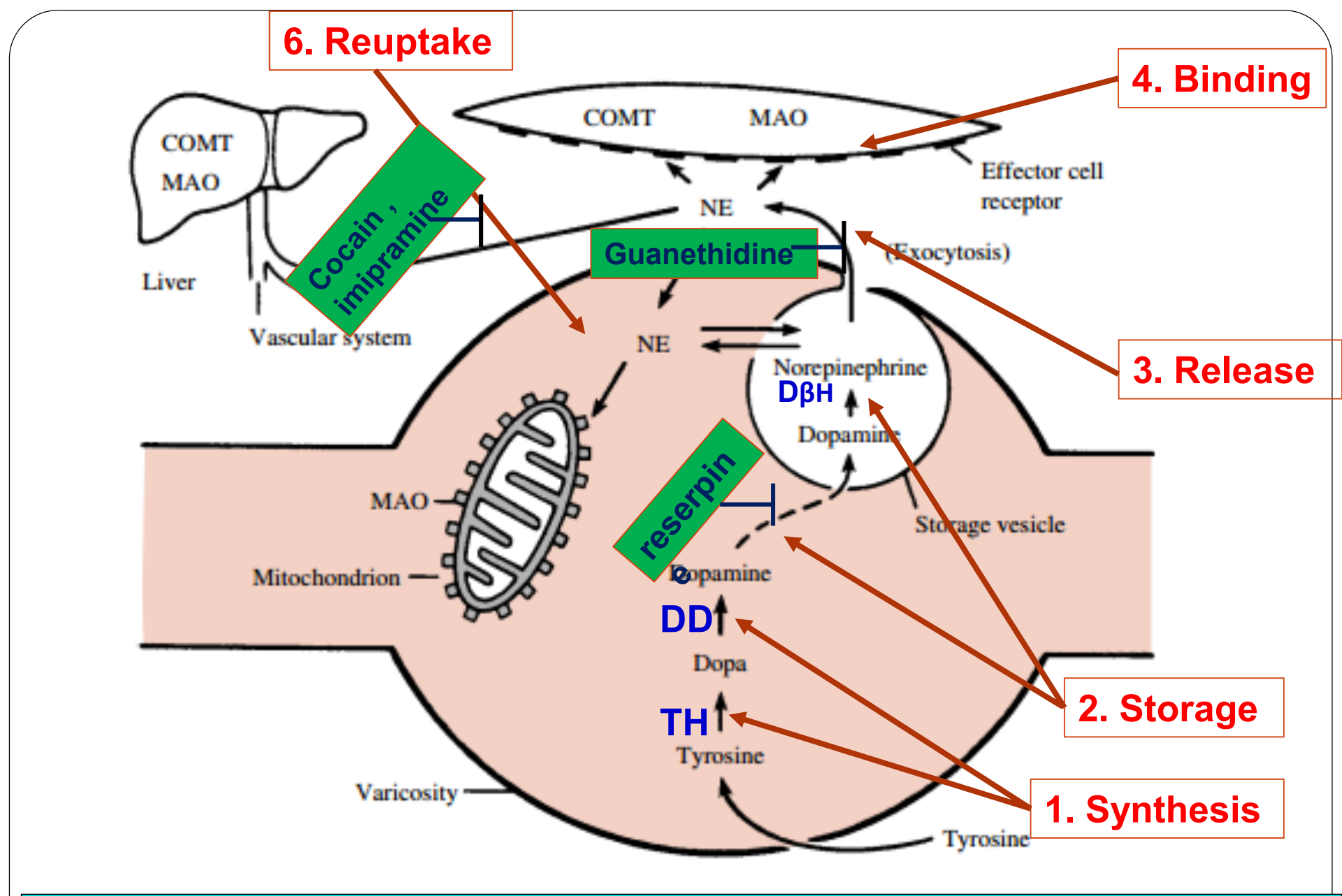


Adrenergic Agonists

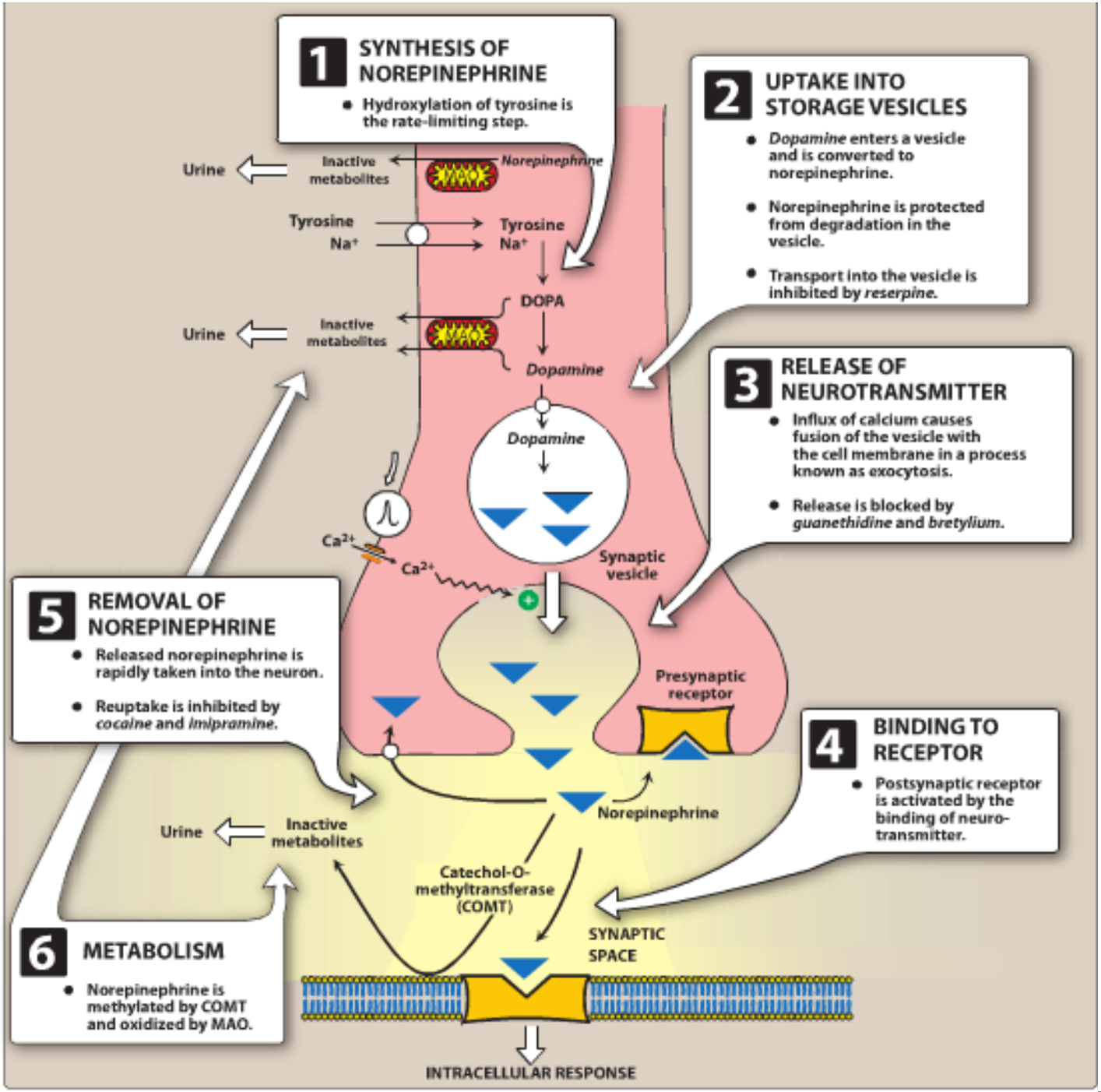
Neurotransmission at adrenergic neurons

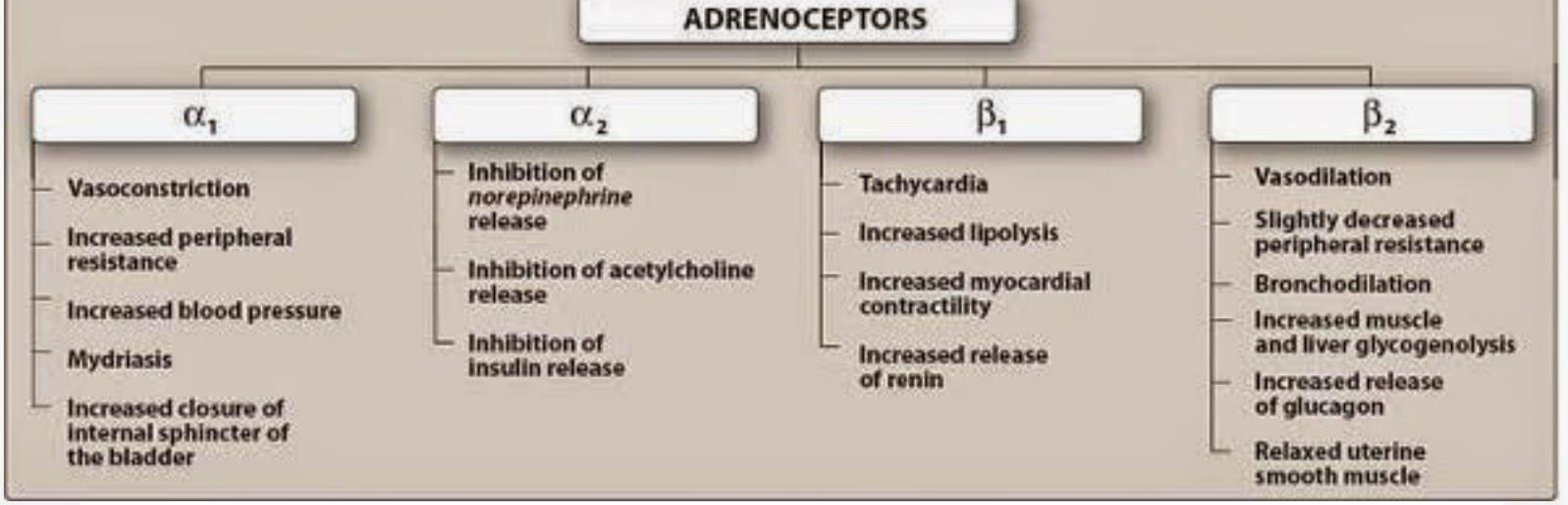
- Neurotransmission takes place at numerous bead-like enlargements called varicosities.

- The process involves five steps:
  1. ***Synthesis***
  2. ***Storage***
  3. ***Release***
  4. ***Receptor binding of NE***
  5. ***Removal of the neurotransmitter***



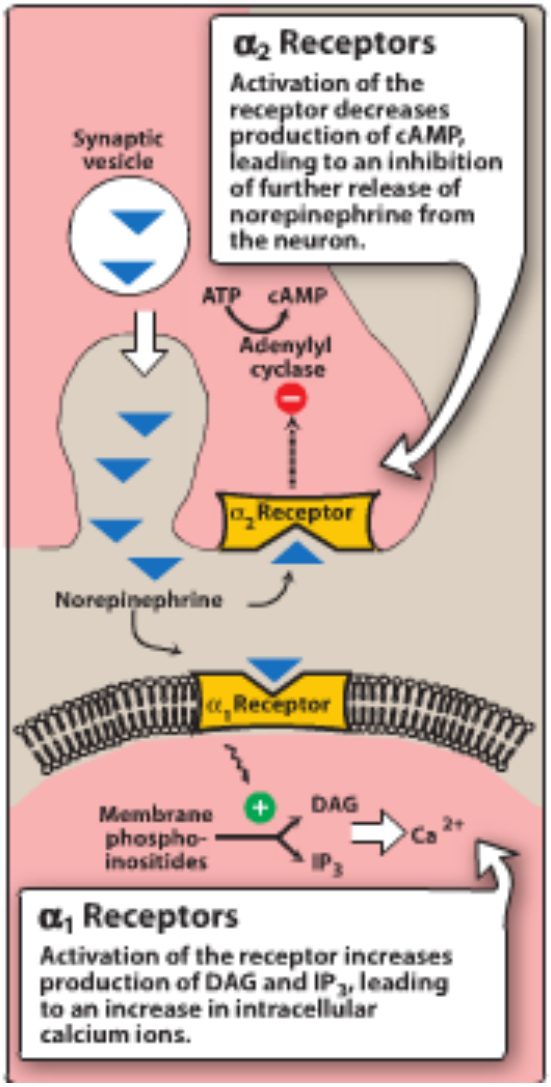
TH: thyroxin hydroxylase, **DD**: dopamine decarboxylase, **DβH**: dopamine β-hydroxylase
COMT: catehol-O-methyltransferase, **MAO**: monoamine oxidase


1 SYNTHESIS OF NOREPINEPHRINE
Hydroxylation of tyrosine is the rate-limiting step.
2 UPTAKE INTO STORAGE VESICLES
Dopamine enters a vesicle and is converted to norepinephrine.
Norepinephrine is protected from degradation in the vesicle.
Transport into the vesicle is inhibited by reserpine.
3 RELEASE OF NEUROTRANSMITTER
Influx of calcium causes fusion of the vesicle with the cell membrane in a process known as exocytosis.
Release is blocked by guanethidine and bretylium.
4 BINDING TO RECEPTOR
Postsynaptic receptor is activated by the binding of neuro-transmitter.
5 REMOVAL OF NOREPINEPHRINE
Released norepinephrine is rapidly taken into the neuron.
Reuptake is inhibited by cocaine and imipramine.
6 METABOLISM
Norepinephrine is methylated by COMT and oxidized by MAO.
Urine
Inactive metabolites
MAO
Norepinephrine
Tyrosine
Na+
DOPA
Dopamine
Ca2+
Synaptic vesicle
Presynaptic receptor
Catechol-O-methyltransferase (COMT)
SYNAPTIC SPACE
INTRACELLULAR RESPONSE



ADRENERGIC RECEPTORS AND SECOND MESSENGERS

Receptor	location	Post receptor mechanism
α_1	Contraction: Most vascular smooth muscle. Pupillary radial muscle Pilomotor smooth muscle, Prostate, Sphincters (Urinary bladder and Anal)	IP3 / DAG cascade (Gq)
α_2	Presynaptic CNS adrenoceptors- Decreases sympathetic flow, GIT (Heterotropic receptors-relaxation), Pancreas	cAMP inhibition (Gi)
B_1	Heart, Juxtaglomerular cells,	cAMP Activation (Gs)
B_2	Relaxation: Smooth muscle of GIT, Urinary bladder, Uterus and Bronchial. Ciliary epithelium, Vascular smooth muscle (Skeletal), Liver	cAMP Activation (Gs)
B_3	Fat cells	cAMP Activation (Gs)



Differences between α1 and α2

	Alpha-1	Alpha-2
Location	Post junctional – blood vessels of skin and mucous membrane, Pilomotor muscle & sweat gland, radial muscles of Iris	Prejunctional
Function	Stimulatory – GU, Vasoconstriction, gland secretion, Gut relaxation, Glycogenolysis	Inhibition of transmitter release, vasoconstriction, decreased central symp. Outflow, platelet aggregation
Agonist	Phenylephrine, Methoxamine	Clonidine
Antagonist	Prazosin	Yohimbine

OVERVIEW

- The adrenergic drugs affect receptors that are stimulated by NE or E.
- These receptors known as **adrenergic receptors** or **adrenoceptors**

- Drugs affecting adrenergic system:
  1. Sympathomimetics (agonists):
     1. **Direct acting agonists**: direct activation of the adrenergic receptor
     2. **Indirect acting agonists**: enhancing release or blocking reuptake of NE
  2. Sympatholytics (antagonists): block the action of the neurotransmitters at the receptors

TISSUE	RECEPTOR TYPE	ACTION	OPPOSING ACTIONS
Heart			
• Sinus and AV	β_1	↑ Automaticity	Cholinergic receptors
• Conduction pathway	β_1	↑ Conduction velocity, automaticity	Cholinergic receptors
• Myofibrils	β_1	↑ Contractility, automaticity	
Vascular smooth muscle	β_2	Vasodilation	α-Adrenergic receptors
Bronchial smooth muscle	β_2	Bronchodilation	Cholinergic receptors
Kidneys	β_1	↑ Renin release	α_1-Adrenergic receptors
Liver	β_2, α_1	↑ Glycogenolysis and gluconeogensis	—

TISSUE	RECEPTOR TYPE	ACTION	OPPOSING ACTIONS
Adipose tissue	β_3	↑ Lipolysis	α_2-Adrenergic receptors
Skeletal muscle	β_2	↑ Increased contractility Potassium uptake; glycogenolysis Dilates arteries to skeletal muscle Tremor	—
Eye-ciliary muscle	β_2	Relaxation	Cholinergic receptors
GI tract	β_2	↓ Motility	Cholinergic receptors
Gall bladder	β_2	Relaxation	Cholinergic receptors
Urinary bladder detrusor muscle	β_2	Relaxation	Cholinergic receptors
Uterus	β_2	Relaxation	Oxytocin

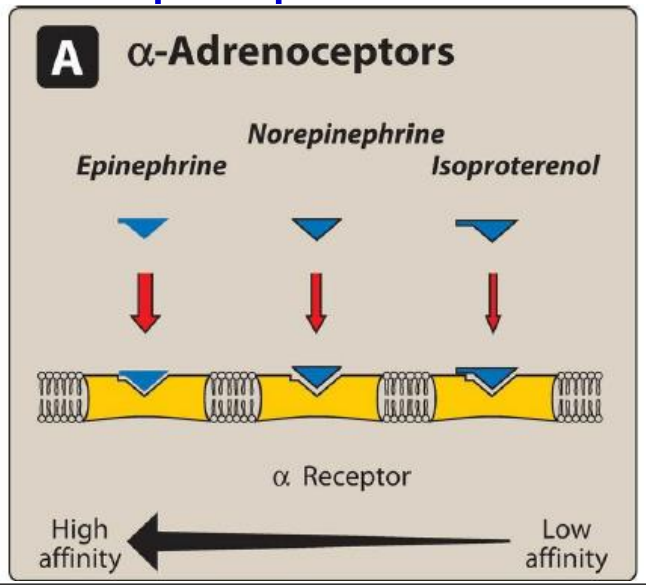
ADRENERGIC AGONISTS

α1 and α2 Receptors:

- The α-adrenoceptors
  - Show weak response to the synthetic agonist isoproterenol
  - They are responsive to the naturally occurring catecholamines epinephrine and norepinephrine.

  - For α receptors, the rank order of potency and affinity

    epinephrine ≥ norepinephrine >> isoproterenol

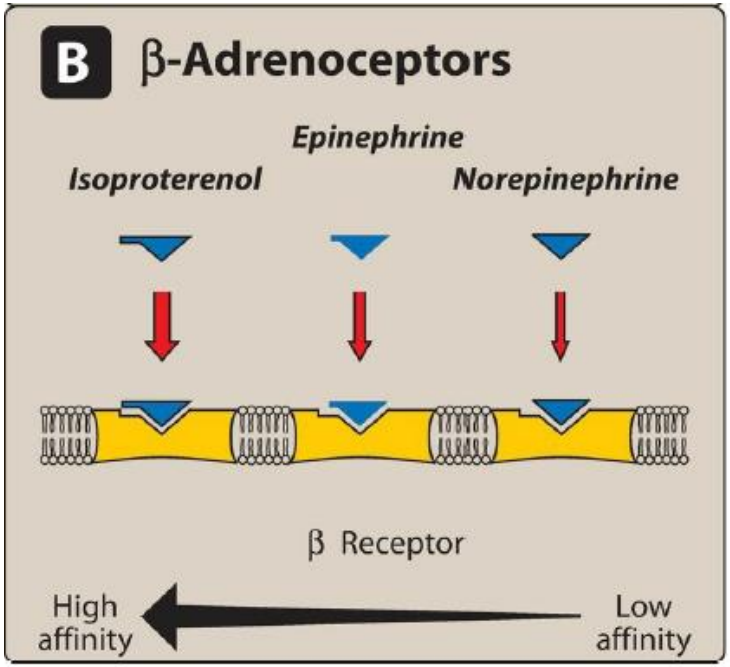


β - adrenoceptors:

- The rank order of potency is

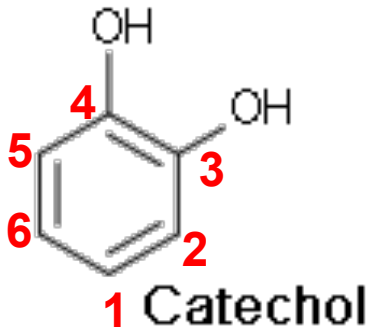
isoproterenol > epinephrine > norepinephrine

- The β-adrenoceptors: β1, β2, and β3
- based on their affinities for adrenergic agonists and antagonists.



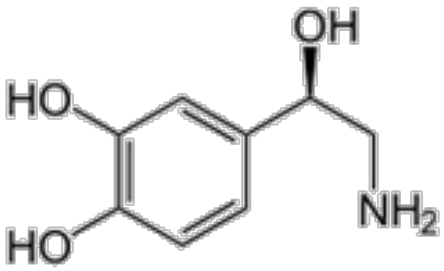
A. Catecholamines

- Catecholamines are sympathomimetic amines that contain the 3,4-dihydroxybenzene group such as:
  1. **Epinephrine**
  2. **Norepinephrine**
  3. **Dopamine**
  4. **Isoproterenol**
  5. **Doputamine**

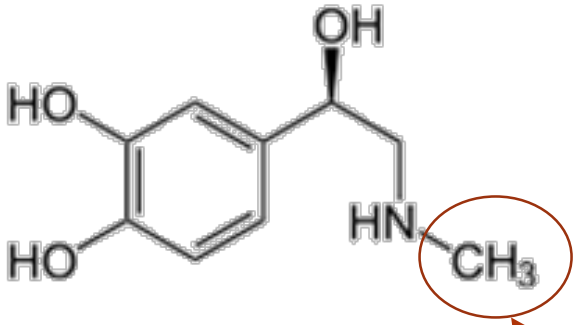


Catecholamines share the following properties:

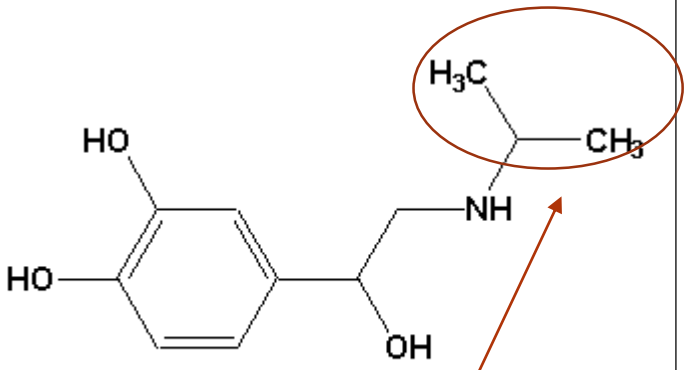
1. High potency: α or β
2. Rapid inactivation: Catecholamines metabolized by:
   - COMT postsynaptically, and in the gut wall
   - MAO intraneuronally, in the liver, and in the gut wall
3. Poor penetration into CNS:
   - Are polar
   - Nevertheless, most of these drugs have some clinical effects (anxiety, tremor, and headaches) that are attributable to action on the CNS.



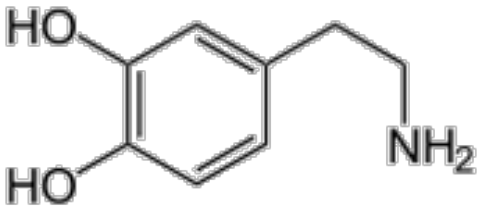
Norepinephrine



Epinephrine



Isoproterenol



Dopamine

Increases the affinity for β receptors

B. Noncatecholamines:

- Lack catechol hydroxyl groups
- These include:
  1. **Phenylephrine**
  2. **Ephedrine**
  3. **Amphetamine**.

- Longer half-lives:
  - Not inactivated by COMT.
  - Poor substrates for MAO
- Increased lipid solubility better access to the CNS.

Note: Ephedrine and amphetamine may act indirectly by causing the release of stored catecholamines.

Mechanism of action of the adrenergic agonists

1. Direct-acting agonists:

- They act directly on α or β receptors
- Examples of direct-acting agonists:
  - Epinephrine
  - Norepinephrine
  - Isoproterenol
  - Phenylephrine

2. Indirect-acting agonists:

- **Block the reuptake** of norepinephrine (uptake blockers) e.g. cocaine, imipramine
- **Cause the release of** norepinephrine from the cytoplasmic pools or vesicles of the adrenergic neuron. e.g. amphetamines

3. Mixed-action agonists:

- Ephedrine
- Pseudoephedrine

- Agonists that have the capacity to:

1. **Stimulate** adrenoceptors directly
2. **Release** norepinephrine from the adrenergic neuron.

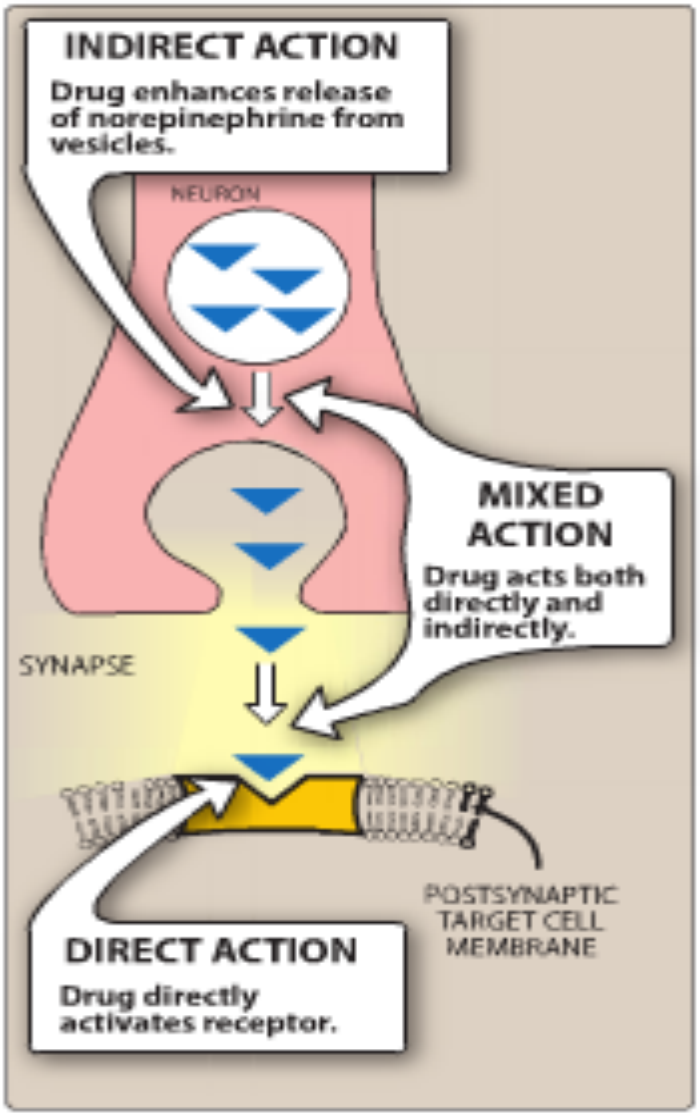


Figure 6.8
Sites of action of direct-, indirect-, and mixed-acting adrenergic agonists.

DIRECT-ACTING ADRENERGIC AGONISTS

- They bind to adrenergic receptors without interacting with the presynaptic neuron.
- The activated receptor initiates synthesis of second messengers and subsequent intracellular signals.
- These agents are widely used clinically.

A. Epinephrine

- Epinephrine is synthesized in the adrenal medulla and released into the bloodstream.

- *Epinephrine* interacts with both α and β receptors.
  - At low doses, β effects (vasodilation) on the vascular system predominate,
  - At high doses, α effects (vasoconstriction) are the strongest.

Adrenergic Agonists: Epinephrine

❖ Cardiovascular:

- Epinephrine is a potent activator of α and ß adrenergic receptors.
  - Prominent cardiovascular effects particularly on blood pressure: potent vasopressor.
  - Systolic pressure increases to a greater extent than diastolic (diastolic pressure may decrease).
- Epinephrine increases blood pressure by:

1. Enhancing cardiac contractility (positive inotropic effect): ß1-receptor effects.
2. Increasing heart rate (positive chronotropic effect): ß1-receptor effects. (↑ Oxygen demand on myocardium).
3. Activates β1 receptors on the kidney to cause renin release.
   - Renin results in production of angiotensin II, a potent vasoconstrictor

Adrenergic Agonists: Epinephrine

- **Respiratory:**
  - Epinephrine is a significant respiratory tract bronchodilator through ß2-receptor activation that mediate smooth muscle relaxation.
  - ß2-receptor activation also decreases mast cell secretion (histamines).
    - This decrease may be beneficial is management of asthma

Adrenergic Agonists: Epinephrine

❖ Hyperglycemia:

- Epinephrine has a significant **hyperglycemic** effect because of:
  1. Increased glycogenolysis in the liver (β2 effect)
  2. Increased release of glucagon (β2 effect)
  3. Decreased release of insulin (α2 effect).

❖ Lipolysis:

- Epinephrine initiates lipolysis through its agonist activity on the β3 receptors of adipose tissue

Epinephrine Therapeutic Use

- **Brochospasm:**
  - Emergency in cases of brochoconstricton resulted in diminished respiratory exchange
    - Selective β2 agonists, such as albuterol, are presently favored because
      - A longer duration of action
      - Minimal cardiac stimulatory effect.

- **Anaphylactic shock:**
  - Drug of choice for Type I hypersensitivity to allergens

Epinephrine Therapeutic Use

- **Glaucoma:**
  - 2% solution topical can reduce intra-ocular pressure
- **In local anesthetics:**
  - Provides vasoconstriction (1:100000):
    1. increases the duration of the local anesthesia.
    2. Minimizes systemic effects
- **Cardiac arrest:**
  - Epinephrine may be used to restore cardiac rhythm in patients with cardiac arrest regardless of the cause.

Epinephrine

- **PK:**
  - Rapid Onset
  - Short Duration (rapid degradation by COMT and MAO)
  - Metabolite in Urine, Orally Ineffective
  - Routes: IM ,IV , Sub Q, Endotracheal Tube, Inhalation

- **Adverse Effect:**
  - CNS: Headache, anxiety, fear, tension, tremor
  - Hemorrhage: marked elevation in BP
  - Cardiac Arrhythmia
  - Pulmonary Edema

- **Interactions:**

❖ **Hyperthyroidism:**

- Increased production of adrenergic receptors on the vasculature leading to a hypersensitive response.
- Epinephrine enhances cardiovascular actions in these patients.
- The dose of epinephrine must be **reduced**.

❖ **Diabetes:**

- Epinephrine increases the release of endogenous stores of glucose.
- In the diabetic, dosages of insulin may have to be **increased**.

B. Norepinephrine

- Norepinephrine is the primary neurotransmitter released by postganglionic neurons of the SANS.
- Norepinephrine (Levophed®) is a potent activator of α and ß1 adrenergic receptors.
- In therapeutic doses, the α- receptor is most affected
  - Potent vasopressor: systolic and diastolic pressure increase.
  - Vasoconstriction α1 receptor effects: precapillary resistance vessels of the skin, kidney, and mucosa.

- **Therapeutic use**: in treatment of shock. (Limited use).

Adrenergic Agonists: Norepinephrine

- In therapeutic doses, the α- receptor is most affected.

1. Cardiovascular actions:

1. Vasoconstriction:

- Increases peripheral resistance: due to intense vasoconstriction of most vascular beds, including the kidney (α1 effect).

- Both systolic and diastolic blood pressures increase.

2. Baroreceptor reflex:

- Negligible cardiac stimulation.

Increased BP

↓

Induces a reflex rise in vagal activity by stimulating the baroreceptors

↓

Reflex bradycardia

- **Effect of atropine pretreatment**

Other Adrenergic Agonists

- **Dopamine:**
  - Immediate precursor of norepinephrine.
  - Dopamine is a CNS neurotransmitter associated with the basal ganglia
  - Activate both α and β adrenoceptors and dopamine receptors.
  - Unique among catecholamines in that dopamine can simultaneously increase:
    - **Myocardial contractility (β1)**
    - **Renal blood flow. (dilates renal arterioles (D1))**
    - **Glomerular filtration rate.**
    - **Sodium excretion.**
    - **Urine output.**

  - **Therapeutic use**: treatment of shock, ↑ BP & kidney perfusion. Continuous IV infusion.

Other Adrenergic Agonists

- **Isoproterenol** (Isuprel®).
  - Direct-acting synthetic catecholamine
  - Activates ß adrenergic receptors (both ß1 - and ß2 -receptor subtypes).
  - Has limited action at alpha adrenergic receptors.
  - Rarely used therapeutically (non-selective).

Other Adrenergic Agonists

- **Dobutamine (Dobutrex®).**
  - Synthatic direct acting catecholamines
  - Pharmacological effects exerted through interaction with **ß1** receptor agonist.
  - No action through dopamine receptors.

- **Therapeutic use.**
  - Short-term management of heart failure following surgery, during acute congestive heart failure, or post-myocardial infarction: ↑ cardiac output, little change in heart rate.
  - Should be used with caution in patients with atrial fibrillation.

CATECHOLAMINES

- Rapid onset of action
- Brief duration of action
- Not administered orally
- Do not penetrate the blood-brain barrier

DRUG	RECEPTOR SPECIFICITY	THERAPEUTIC USES
Epinephrine	α_1, α_2 β_1, β_2	Acute asthma Anaphylactic shock In local anesthetics to increase duration of action
Norepinephrine	α_1, α_2 β_1	Treatment of shock
Isoproterenol	β_1, β_2	As a cardiac stimulant
Dopamine	Dopaminergic α_1, β_1	Treatment of shock Treatment of congestive heart failure Raise blood pressure
Dobutamine	β_1	Treatment of acute heart failure

α1 Selective Adrenergic Agonists

- α1 adrenergic agonists are used clinically in management of hypotension / shock.
  - **Phenylephrine** and **Methoxamine** are direct-acting vasoconstrictors drugs.
  - Activate **α1 adrenergic receptors in vascular smooth** muscle producing vasoconstriction
  - Peripheral vascular resistance is increased.
  - Blood **pressure may be increased**, causing a reflex reduction heart rate **(Bradycardia).**

α1 Selective Adrenergic Agonists

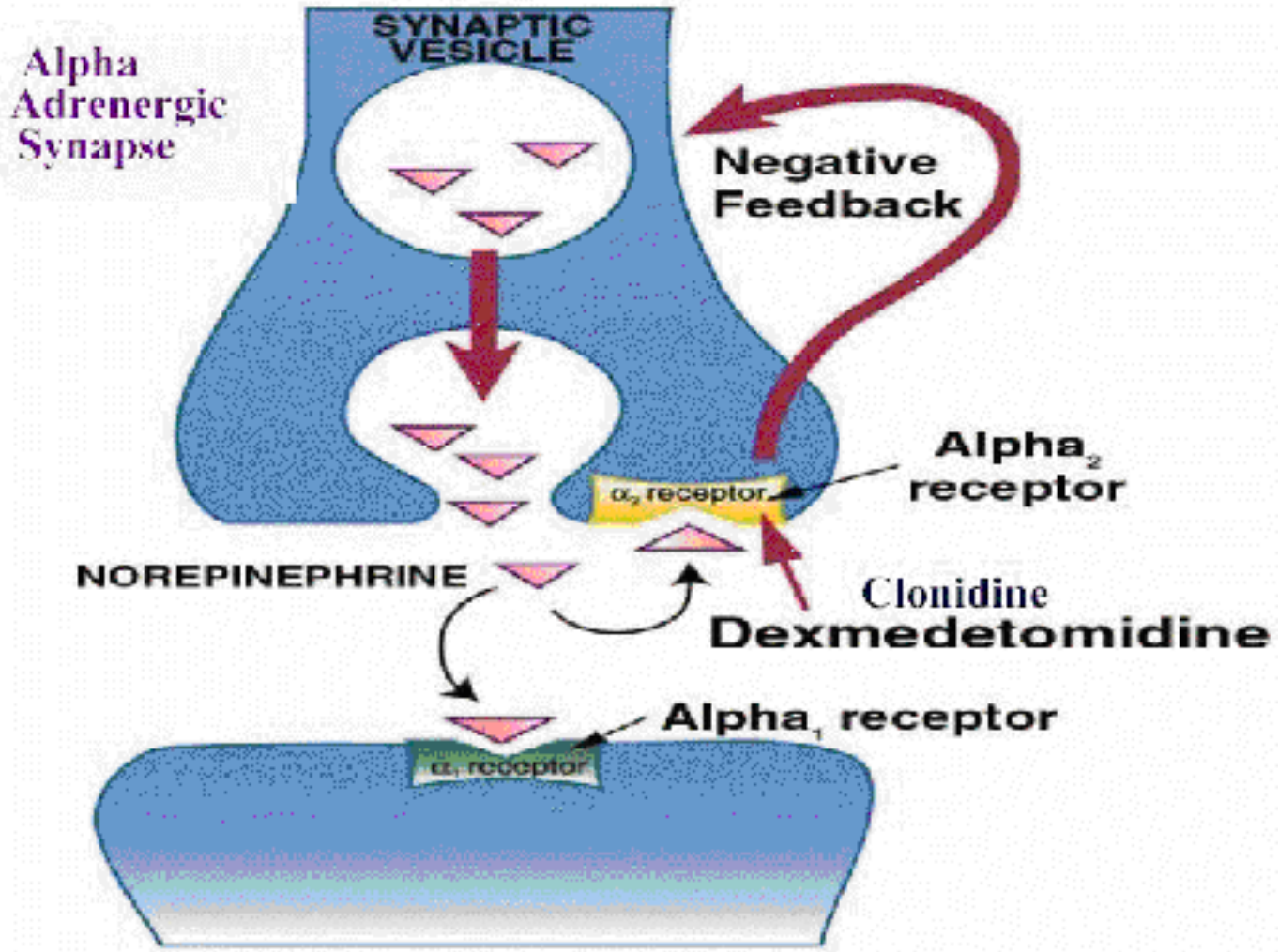
- **Phenylephrine**
  - No effect on heart
  - Used to treat hypotension in hospitalized patients
  - Topically and orally used as nasal decongestant (replaced pseudoephedrine)
  - Ophthalmic solution for Mydriasis

- **Methoxamine**:
  - To treat hypotension during surgery involving halothane anesthesia.
  - Does not trigger cardiac arrhythmia sensitized by general anesthesia.

α2 Selective Adrenergic **Agonists**

- α2 selective adrenergic agonists are used to treat **essential hypertension**.
- Mechanism of action:
  - Activation of central α_2 adrenergic receptors at cardiovascular control centers.
  - Activation decreases sympathetic outflow, reducing sympathetic vascular tone.

Central-acting Adrenergic: Clonidine



α2 Selective Adrenergic Agonists

- **Clonidine (Catapres®).**
- Therapeutic use: in treating essential hypertension.
  - Clonidine acts in the brain at pre-synaptic α2 receptors by inhibiting adrenergic outflow from the brainstem.
  - Inhibition of sympathetic outflow results in a decrease in blood pressure.
  - Clonidine reduces cardiac output (by reducing both stroke volume and heart rate) and peripheral resistance.
  - A prolonged hypotensive response results.

α2 Selective Adrenergic Agonists

- Clonidine (Catapres®): **adverse effects:**
  - Lethargy
  - Dry mouth (xerostomia).
  - Sedation.
  - Sexual dysfunction.

- **Other drugs:**
  - **Guanabenz (Wytensin®).**
  - **Guanfacine.**
  - **Alpha-methyl DOPA- (methyldopa (Aldomet®))** metabolically converted to alpha-methyl norepinephrine.
    - A favorite during pregnancy.
- **Same mechanism, same indication, same side effects.**

ß2 Selective Adrenergic **Agonists**



Albuterol (Ventolin®) and Terbutaline

- Short β2 agonists (less than 3 hours)
- Effective in managing asthma (metered-dose bronchodilator)
- Orally active and metabolized more slowly compared to catecholamines
- In asthma, pulmonary ß2 receptors are targeted by drug administration by inhalation
- Pulmonary ß2 receptors result in smooth-muscle relaxation and bronchodilation
- Decrease histamine and leukotriene release from lung mast cells

ß2 Selective Adrenergic Agonists

- Adverse effects-agonists.
  - Excessive cardiovascular stimulation.
  - Skeletal muscle tremor (tolerance develops, unknown mechanism) due to ß2 adrenergic receptor activation.
  - Over-usage may be a factor in morbidity and mortality in asthmatics.

ß2 Selective Adrenergic Agonists

Salmeterol and formoterol

- Are long-acting β2-agonists (LABAs) that are bronchodilators.
- Duration: 12 hours (dried powder inhaler)
- Has a delayed onset of action.
- Used to treat nocturnal asthma in symptomatic patients taking other asthma medications
- Inhaled β2-receptor agonists should not be used in excess.

ß2 Agonist: Uterine Relaxant

Ritodrine (Yutopar®), Terbutaline (Brethine®)

- ß2 adrenergic receptor-selective: developed as a uterine relaxant.
- May be administered by I.V. In certain patients for arresting premature labor; If successful, oral therapy may be started.
- ß2 adrenergic receptor-selective agonists may not improve perinatal mortality and may increase maternal morbidity.
- In women being treated for premature labor, may have pulmonary edema.

Ephedrine (No longer recommended)

- **Mixed-acting adrenergic agonist**
- Produced by various plants (Ma-huang)
- Noncatechol with good oral bioavailabilty
- Long acting (poor substrates for COMT and MAO)
- Releases catecholamines from nerve terminal
- Some direct stimulation of α and β receptors
- Penetrates brain to produce CNS stimulation

Indirect acting Sympathomimetic

Amphetamine (Global®)

- **Non-catechol**
- **Good oral bioavailability**
- **CNS stimulant - more than ephedrine**
- **Peripheral actions mainly through release of catecholamines from nerve terminals**

Tyramine

- Releases NE from nerve terminals
- Found in fermented foods (e.g. wine, cheese)
- Normally metabolized by MAO
- May produce hypertensive crisis in patients taking MAO inhibitors

Cocaine

- Rapidly enters the CNS
  - Like amphetamine
  - Shorter acting
  - More intense than amphetamine
- Blocks **NE reuptake** into nerve terminal
- Peripheral sympathomimetic effects prominent
- Smoked, snorted and injected for rapid onset

NONCATECHOL-AMINES

Compared to catecholamines:

- **Longer duration of action**
- **All can be administered orally**

DRUG	RECEPTOR SPECIFICITY	THERAPEUTIC USES
Oxymetazoline	α_1	As a nasal decongestant
Phenylephrine	α_1	As a nasal decongestant Raise blood pressure Treatment of paroxysmal supraventricular tachycardia
Methoxamine	α_1	Treatment of supraventricular tachycardia
Clonidine	α_2	Treatment of hypertension
Albuterol *Terbutaline*	β_2	Treatment of bronchospasm (short acting)
Salmeterol *Formoterol*	β_2	Treatment of bronchospasm (long acting)
Amphetamine	α, β, CNS	As a CNS stimulant in treatment of children with attention deficit syndrome, narcolepsy, and for appetite control
Ephedrine *Pseudoephedrine*	α, β, CNS	As a nasal decongestant Raise blood pressure

Adrenergic Antagonists

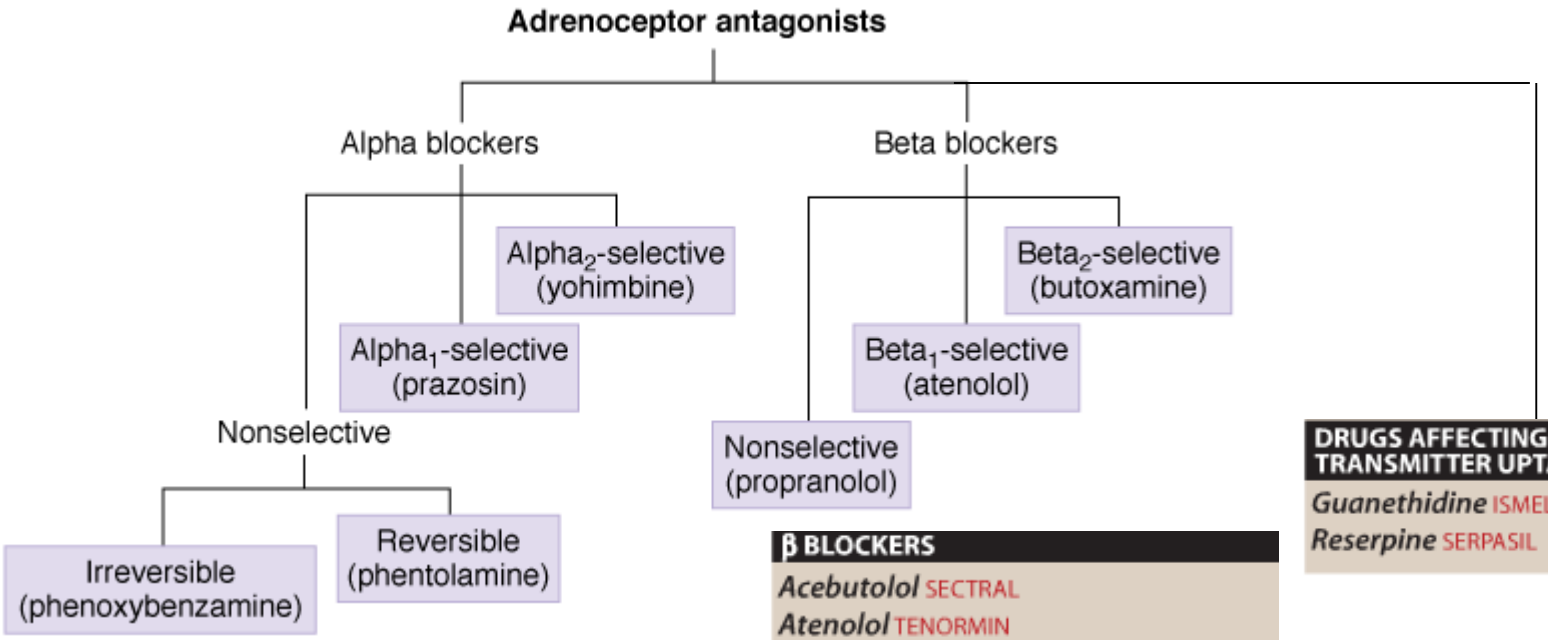
Adrenergic blockers
Sympatholytics

Their action?

- These drugs act by attaching to the receptor, thus preventing its activation by endogenous catecholamine.

- Adrenergic antagonists are classified according to their relative affinities for α or β receptors in the peripheral nervous system:
  - α - blockers
  - β - blockers

Classification



α BLOCKERS

Alfuzosin UROXATRAL
Doxazosin CARDURA
Phenoxybenzamine DIBENZYLINE
Phentolamine REGITINE
Prazosin MINIPRESS
Tamsulosin FLOMAX
Terazosin HYTRIN
Yohimbine YOCON

β BLOCKERS

Acebutolol SECTRAL
Atenolol TENORMIN
Betaxolol BETOPTIC-S, KERLONE
Bisoprolol ZEBETA
Carteolol CARTROL
Carvedilol COREG, COREG CR
Esmolol BREVIBLOC
Labetalol TRANDATE
Metoprolol LOPRESSOR, TOPROL-XL
Nadolol CORGARD
Nebivolol BYSTOLIC
Penbutolol LEVATOL
Pindolol VISKEN
Propranolol INDERAL LA, INNOPRAN XL
Timolol BETIMOL, ISTALOL, TIMOPTIC

DRUGS AFFECTING NEURO-TRANSMITTER UPTAKE OR RELEASE

Guanethidine ISMELIN
Reserpine SERPASIL

Alpha-adrenergic Receptor Antagonists

- **α** adrenoceptors blockers profoundly affect **blood pressure**.
- Blockade of these receptors reduces the sympathetic tone of the blood vessels, resulting in decreased peripheral vascular resistance.
- This induces a reflex tachycardia.
- β receptors are not affected by α blockers.

Therapeutic Uses for Alpha-adrenergic Blocking

Drugs: Phenoxybenzamine & Phentolamine

- **Phenoxybenzamine**: irreversible inhibitor (last about 24 hours)
- **Phentolamine**: reversible inhibitor (last for 4 hours)
- Comparable potency to inhibit α1 and α2-adrenergic receptors
- Used for Treatment of
  - Hypertension caused by pheochromocytoma.
    - Pheochromocytomas are adrenal medullary tumors secreting into the circulation large amounts of catecholamines.
  - Benign Prostatic Hypertrophy (BPH)

α1-adrenergic Blocking Drugs: Prazosin, Terazosin & Doxazosin.

- **Selective α1** adrenergic receptor blockers.
  - Reduction in peripheral vascular resistance.
  - Decrease preload: no significant increase cardiac output or heart rate.
  - Possible CNS effect: suppression of sympathetic outflow.

- **Therapeutic uses:**
  - Essential hypertension.
  - Congestive heart failure (arteriolar and venular dilatation).
  - Benign Prostatic Hypertrophy (BPH): (Tamsulosin and alfuzocin)
  - These drugs improve lipid profiles and glucose metabolism in hypertensive patients

α1-adrenergic Blocking Drugs: Prazosin, Terazosin & Doxazosin.

- **Cardiovascular effects.**
  - α1-adrenergic receptor blockade inhibits vasoconstriction caused by endogenous catecholamines.
  - α1-adrenergic receptor blockers cause an arteriolar and venular vasodilation ↓ (peripheral resistance) that decreases blood pressure.
  - Hypotensive responses trigger baroreceptor reflexes that act to increase:
    - Heart rate (Reflex tachycardia).
    - Cardiac output.
    - Fluid retention.

α1-adrenergic Blocking Drugs: Prazosin, Terazosin & Doxazosin.

- **Adverse effects:**
  - Significant orthostatic hypotension following first-dose (use ¼ Dose at Bed time)
  - Additive effect with Diuretics or ß Blockers (reduce Dose)
  - Male Sexual dysfunction
  - Dizziness, lack of energy, nasal congestion, headache

β-ADRENERGIC BLOCKING AGENTS

ß-Adrenergic Antagonists

- Competitive antagonists.
- Non-selective: β1 and β2 receptors as **Propranolol**
- Cardioselective β antagonists primarily block β1 receptors as **Metoprolol** and **atenolol**
- ß receptor blockers exert most of their therapeutic actions on the cardiovascular system

- The names of all β blockers end in "-olol" except for labetalol and carvedilol.

ß-Adrenergic Antagonists

- **Cardiovascular effect:**
  - Decrease heart rate (-ve chronotropic).
  - Reduce contractility (-ve iononotropic).
  - Reduce myocardial oxygen demand (treatment of angina pectoris).

- **Antihypertensive effects**.
  - ß receptor blockers reduce blood pressure in patients with hypertension (not in normal subjects).

- The long-term reduction in peripheral resistance, in addition to a reduction in cardiac output is the main benefit of ß-blockers.

- **Pulmonary effects:**
  - In asthmatic patients or patients with COPD (chronic obstructive pulmonary disease) ß receptor blockers cause dangerous bronchiolar constriction.
  - This effect is particularly prominent with nonselective ß receptor blockers, such as Propranolol.

- **Metabolic actions:**
  - β Blockade leads to decreased:
    - Glycogenolysis
    - glucagon secretion.
  - Patient with type 1 diabetes is to be given propranolol, very careful monitoring of blood glucose is essential, because pronounced hypoglycemia may occur after insulin injection.
  - β Blockers also attenuate the normal physiologic response to hypoglycemia.
  - Because of its ability to suppress glycogenolysis and mask tachycardia, propranolol must be used with caution by diabetic patients."

Nonselective-ß adrenergic Receptor Antagonists: Propranolol

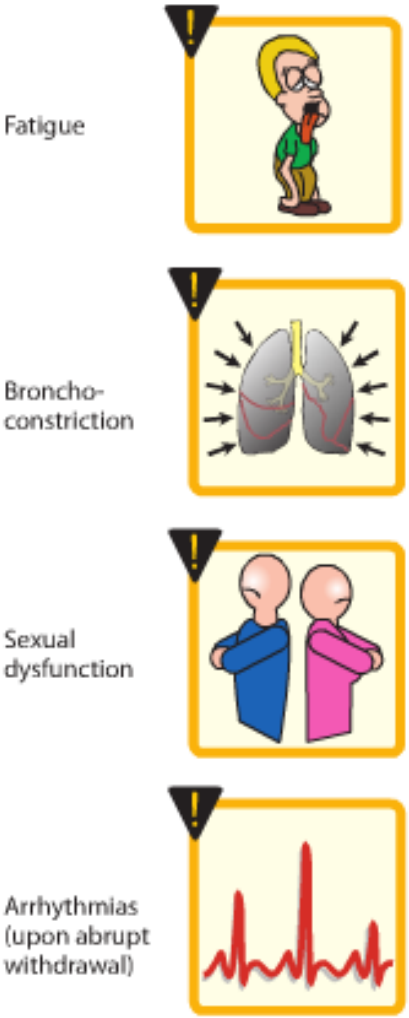
- Lipophillic nonselective ß adrenergic receptor antagonist.

- **Effective in treating.**
  - Essential hypertension.
  - Angina, Myocardial Infarction.
  - Certain arrhythmias.
  - Glaucoma: ↓ Secretion of Aq. Humor (chronic).
  - Migraine: Chronic, reduce incidence and severity.
  - Hyperthyroidism: blunt sympathetic Stimulation.

Propranolol Adverse Events

- Bronchoconstriction
- Arrhythmias (if treatment is stopped quickly)
- Sexual Impairment
- Disturbances in metabolism

Adverse effects:



The reasons for this are not clear and may be independent of β-receptor blockade.

The β blockers must be tapered of gradually for at least a few weeks.

Nonselective-ß adrenergic Receptor Antagonists:

- **Nadolol**.
  - Long half-life, long duration.
  - May accumulate in patient with renal dysfunction.

- **Timolol:**
  - Ocular use: treatment of glaucoma.
  - ↓ Secretion of Aq. Humor (chronic).
  - Systemic effects common after ocular application.
    - bradycardia and hypotension

Selective- ß1 Receptor Antagonists

Selective- ß1 Receptor Antagonists

- **Metoprolol** (Lopressor®).
  - Effective in treating essential hypertension.
  - Effective in treating symptoms of coronary artery disease.
  - Lower risk than non-selective in patients with patients with impaired pulmonary function or Diabetic hypertensive patients.

Selective- ß1 Receptor Antagonists

- **Esmolol**
  - Very short duration of action (9 minutes).
  - I.V. Administration.
  - Effective in rapid termination of supraventricular arrhythmias.
  - Beneficial in management of intra-operative adverse BP/heart rate increases associated with noxious stimulation (e.g. During tracheal intubation).

Selective- ß1 Receptor Antagonists

- **Atenolol**
  - Most **cardio-selective** ß1 adrenergic receptor antagonist.
  - Used in treating hypertension.
  - Somewhat longer action compared to metoprolol.

Therapeutic use:

1. The cardioselective β blockers are useful in hypertensive patients with impaired pulmonary function.
   - Because these drugs have less effect on peripheral vascular β2 receptors.

2. Cardioselective β blockers are useful in diabetic hypertensive patients.

Antagonist and partial agonist

Pindolol and Acebutolol

- Intrinsic sympathomimetic activity

- Most useful in hypertensive patients with moderate bradycardia because:
  - Further ↓ in heart rate is less pronounced with these drugs
- Carbohydrate and lipid metabolism less affected than other β-blockers. Thus may be an option in treating diabetics
- Due to ISA, cannot use in patients with arrhythmia

Therapeutic use :

1. Hypertensive patients with moderate bradycardia: because a further decrease in heart rate is less pronounced with these drugs.

2. Diabetic hypertensive patients: carbohydrate metabolism is less affected.

Antagonists of both α and β adrenoceptors: Labetalol and carvedilol

Actions:

- α1-blocking actions: produce peripheral vasodilation, thereby reducing blood pressure.
- They do not alter serum lipid or blood glucose levels.

- Carvedilol also decreases lipid peroxidation and vascular wall thickening, effects that have benefit in heart failure.

Therapeutic use in:

1. Emergency hypertension and Pregnancy induced Hypertension (labetalol)
2. Heart failure (carvidiol)

3. Adverse effects: associated with α1 blockade

- Orthostatic hypotension
- Dizziness

DRUG	RECEPTOR SPECFICITY	THERAPEUTIC USES
Propranolol	β_1, β_2	Hypertension Migraine Hyperthyroidism Angina pectoris Myocardial infarction
Nadolol	β_1, β_2	Hypertension
Timolol	β_1, β_2	Glaucoma, hypertension
Acebutolol[1] *Atenolol* *Esmolol* *Metoprolol*	β_1	Hypertension
Nebivolol	β_1, NO ↑	Hypertension
Pindolol[1]	β_1, β_2	Hypertension
Carvedilol *Labetalol*	$\alpha_1, \beta_1, \beta_2$	Hypertension Congestive heart failure

Figure 7.13
Summary of β-adrenergic antagonists. NO = nitric oxide. [1]*Acebutolol* and *pindolol* are partial agonists.

DRUGS AFFECTING NEUROTRANSMITTER RELEASE OR UPTAKE

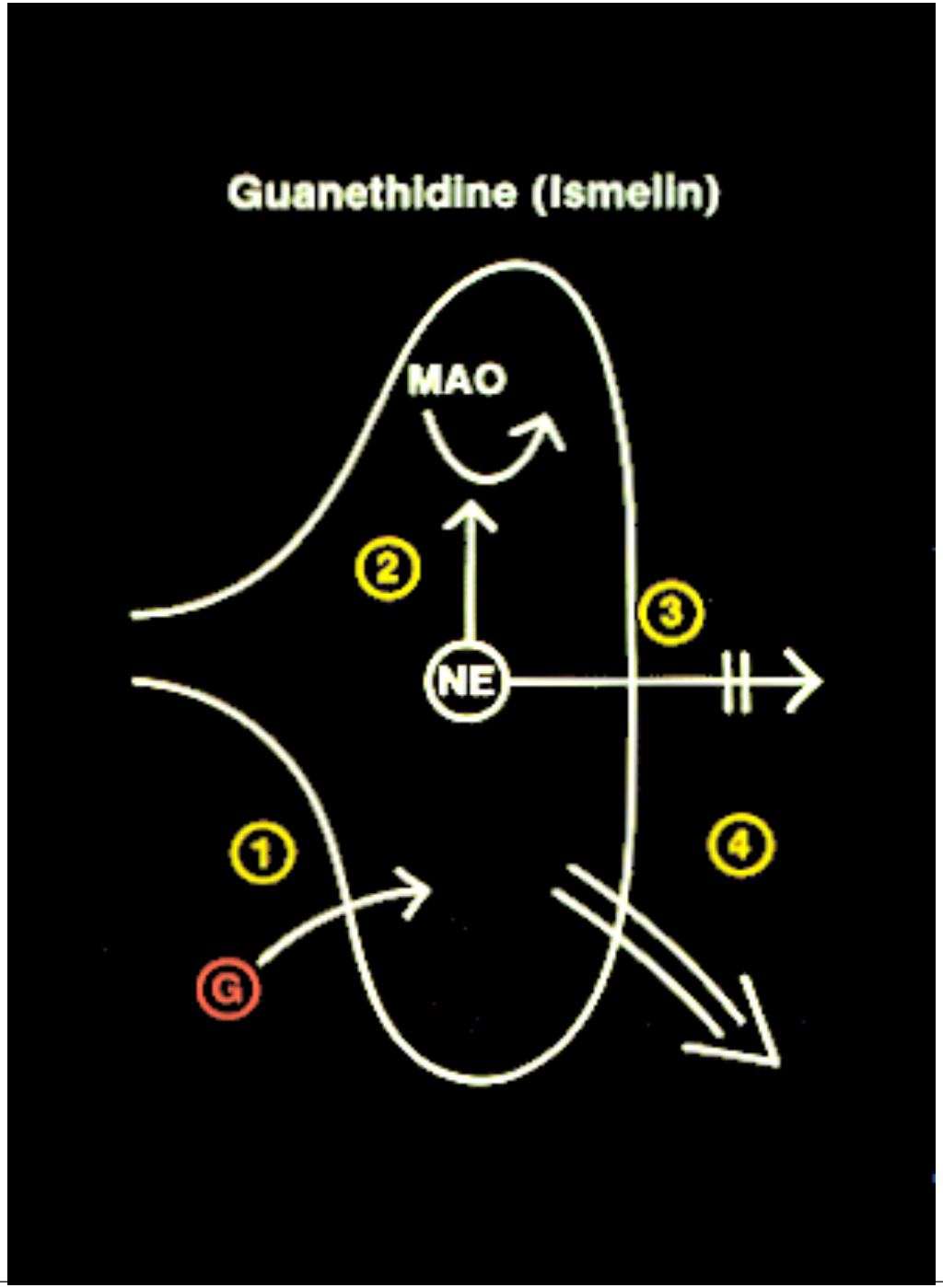
- **Prevent normal physiologic release or storage of NE**
- **These agents are seldom used therapeutically**
  A. **Reserpine**
  B. **Guanethidine**

Guanethidine

Action:

- Enters the neuron via the amine pump
- Releases NE intraneuronally
- Inhibits NE release from sympathetic nerves (anti-release action)
- Depletes NE in peripheral tissues
- Does not penetrate brain

- **Use**: Severe hypertension


Guanethidine (Ismelin)
MAO
2
3
NE
1
4
G

Reserpine

- Inhibits transport of dopamine and NE into the storage granule
- Depletes NE from peripheral and central neurons
- Major side-effects: Depression and gut problems